LESSON PLAN · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

Lesson 8: Searching Lists

Module 3 | 45 minutes | List search | Grades 3–5 · ages 9–11

Objective: Check whether a list contains something and use it to make decisions, like a password checker or an inventory.

Success criteria

- I can check whether a list contains something.
- I can use it in if/else decisions.
- I can build an inventory or password check.

Key vocabulary

- **contains**: True if the item is somewhere in the list.
- **inventory**: A list of things the player has collected.
- **password**: A secret word that lets you in.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Bouncer game (5 min): one learner holds a guest list; others say names and the bouncer answers “in” or “out”.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	The Secret Door: learners follow the steps.
14 min	**Perform**	Code Lab: secret door, Check the bag
9 min	**Probe**	Code Lab: inventory, activity
4 min	**Wrap-up**	Exit ticket: Write the condition that checks if the bag has a key.

Unplugged warm-up

Bouncer game (5 min): one learner holds a guest list; others say names and the bouncer answers “in” or “out”.

Teaching the idea

List membership (contains) is a Boolean usable in if/else. It enables passwords, inventories and keyword checks. Scratch's contains is case-insensitive and matches whole items.

Analogy: A guest list at a party: the guard checks if your name is on it.

Live demo: Live test the password program with right, wrong and CAPITAL answers.

Questions to ask

Ask	Listen for
What does contains report?	True or false.
How would you check something is NOT in the list?	not <list contains ...>.
Where else are lists searched?	Logins, spell-checkers, shopping apps.

Differentiation

Support
Provide the list and ask; learners build the if/else.

Stretch
Count wrong tries with a variable and lock after 3.

Common misconceptions

Misconception	How to fix it
contains finds part of a word	It matches whole items only.
You must check each item with its own if	contains searches the whole list at once.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Write the condition that checks if the bag has a key.

Homework
Write a list of 5 words that a "banned words" filter could check for in a kind chat app.

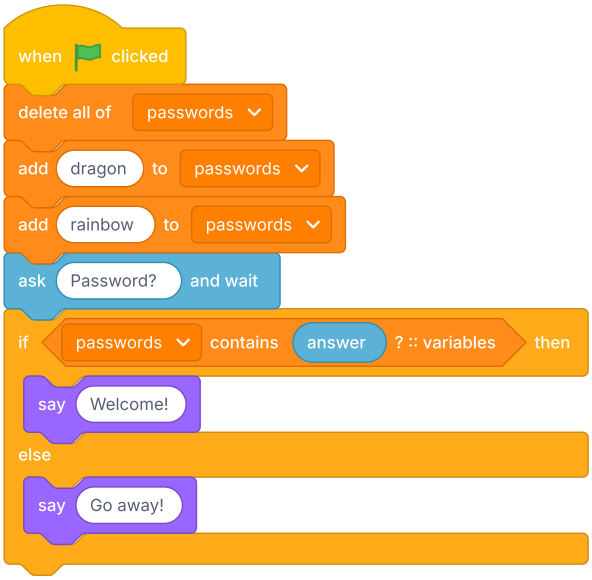
Answer key — Kitabash activities

1. Code Lab: secret door (perform · code lab)

Goals checked: Right password works · Wrong password is refused · Has both passwords

One solution:

Guard



2. Check the bag (perform · multiple choice)

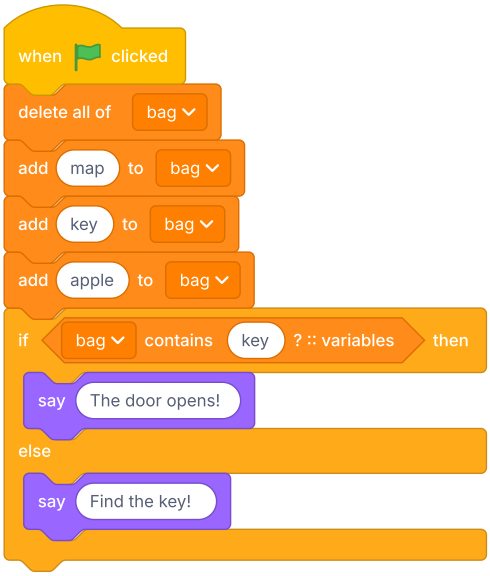
Answer: bag contains key

3. Code Lab: inventory (probe · code lab)

Goals checked: Bag holds 3 things · Checks the bag · Opens the door

One solution:

Explorer



4. Explain your thinking (probe · free response)

Look for: list, add, contains. **Example:** In my castle game, when you touch a sword it is added to the bag list. At the dragon, the game checks if the bag contains sword. If it does you win, and if not the dragon chases you away.

Answer key — worksheet

1. **Key words:** contains = True if the item is somewhere in the list. · inventory = A list of things the player has collected. · password = A secret word that lets you in.
2. **Plan your code:** Goals: Right password works; Wrong password is refused; Has both passwords.
3. **Check the bag:** A. bag contains key — key is in the bag; sword is not; apple is.
4. **Plan your code:** Goals: Bag holds 3 things; Checks the bag; Opens the door.
5. **Explain your thinking:** In my castle game, when you touch a sword it is added to the bag list. At the dragon, the game checks if the bag contains sword. If it does you win, and if not the dragon chases you away.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.